

Microsoft Windows [Version 10.0.26200.7705]
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C:\Users\Varshini S>mysql -u root -p

Enter password: ****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 40

Server version: 9.3.0 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;

Database
bank_simulation_system
college
information_schema
mysql
performance_schema
sakila
sys
world

8 rows in set (0.107 sec)

mysql> use bank_simulation_systems;

Database changed

mysql> CREATE TABLE branches (

-> branch_id INT PRIMARY KEY AUTO_INCREMENT,

-> branch_name VARCHAR(100) NOT NULL,

-> city VARCHAR(50),

-> IFSC_code VARCHAR(20) UNIQUE

->);

Query OK, 0 rows affected (0.769 sec)

mysql> CREATE TABLE customers (

-> customer_id INT PRIMARY KEY AUTO_INCREMENT,

-> customer_name VARCHAR(100) NOT NULL,

-> email VARCHAR(100) UNIQUE,

-> phone VARCHAR(15) UNIQUE,

-> address TEXT,

-> created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

->);

Query OK, 0 rows affected (0.654 sec)

```
mysql> CREATE TABLE accounts (
->     account_id INT PRIMARY KEY AUTO_INCREMENT,
->     account_number VARCHAR(20) UNIQUE NOT NULL,
->     customer_id INT,
->     branch_id INT,
->     account_type ENUM('Savings', 'Current'),
->     balance DECIMAL(10,2) DEFAULT 0,
->     status ENUM('Active', 'Closed') DEFAULT 'Active',
->     opened_date DATE,
->     FOREIGN KEY (customer_id) REFERENCES customers(customer_id),
->     FOREIGN KEY (branch_id) REFERENCES branches(branch_id)
-> );
Query OK, 0 rows affected (0.861 sec)
```

```
mysql> INSERT INTO branches (branch_name, city, IFSC_code) VALUES
-> ('Main Branch', 'Bangalore', 'IFSC001'),
-> ('City Branch', 'Mysore', 'IFSC002'),
-> ('Tech Park Branch', 'Chennai', 'IFSC003');
Query OK, 3 rows affected (0.299 sec)
Records: 3 Duplicates: 0 Warnings: 0

mysql> INSERT INTO customers (customer_name, email, phone, address) VALUES
-> ('Suresh Kumar', 'suresh@gmail.com', '900000001', 'Bangalore'),
-> ('Ananya Rao', 'ananya@gmail.com', '900000002', 'Mysore'),
-> ('Ravi Teja', 'ravi@gmail.com', '900000003', 'Chennai'),
-> ('Sneha Patel', 'sneha@gmail.com', '900000004', 'Mumbai'),
-> ('Suman Das', 'suman@gmail.com', '900000005', 'Kolkata'),
-> ('Kiran Shah', 'kiran@gmail.com', '900000006', 'Delhi'),
-> ('Sanjay Verma', 'sanjay@gmail.com', '900000007', 'Pune'),
-> ('Swathi N', 'swathi@gmail.com', '900000008', 'Bangalore'),
-> ('Amit Jain', 'amit@gmail.com', '900000009', 'Jaipur'),
-> ('Sonia Malhotra', 'sonia@gmail.com', '900000010', 'Delhi');
Query OK, 10 rows affected (0.220 sec)
Records: 10 Duplicates: 0 Warnings: 0

mysql> INSERT INTO accounts (account_number, customer_id, branch_id, account_type, balance, opened_date) VALUES
-> ('ACC1001', 1, 1, 'Savings', 25000, '2024-01-10'),
-> ('ACC1002', 1, 2, 'Current', 40000, '2024-02-15'),
-> ('ACC1003', 2, 1, 'Savings', 15000, '2023-12-01'),
-> ('ACC1004', 3, 3, 'Savings', 50000, '2024-03-20'),
-> ('ACC1005', 4, 2, 'Current', 60000, '2022-05-10'),
-> ('ACC1006', 5, 1, 'Savings', 8000, '2024-04-01'),
-> ('ACC1007', 6, 3, 'Savings', 120000, '2023-11-11'),
-> ('ACC1008', 7, 2, 'Current', 30000, '2024-01-25'),
-> ('ACC1009', 8, 1, 'Savings', 45000, '2024-02-05'),
-> ('ACC1010', 9, 3, 'Savings', 9000, '2023-06-15'),
-> ('ACC1011', 10, 1, 'Current', 70000, '2024-03-12'),
-> ('ACC1012', 2, 2, 'Savings', 20000, '2024-04-10'),
-> ('ACC1013', 3, 1, 'Current', 10000, '2023-09-09'),
-> ('ACC1014', 4, 3, 'Savings', 55000, '2024-01-01'),
-> ('ACC1015', 5, 2, 'Savings', 18000, '2024-02-20');
Query OK, 15 rows affected (0.101 sec)
Records: 15 Duplicates: 0 Warnings: 0
```

```
+-----+
| Tables_in_bank_simulation_systems |
+-----+
| accounts                          |
| branches                         |
| customers                        |
+-----+
3 rows in set (0.526 sec)

mysql> CREATE TABLE transactions (
->     transaction_id INT PRIMARY KEY AUTO_INCREMENT,
->     account_id INT,
->     transaction_type ENUM('Deposit', 'Withdrawal', 'Transfer'),
->     amount DECIMAL(10,2),
->     transaction_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
->     reference_account VARCHAR(20),
->     FOREIGN KEY (account_id) REFERENCES accounts(account_id)
-> );
Query OK, 0 rows affected (0.691 sec)
```

```
mysql> INSERT INTO transactions (account_id, transaction_type, amount) VALUES
-> (1, 'Deposit', 5000), (1, 'Withdrawal', 2000),
-> (2, 'Deposit', 10000), (3, 'Withdrawal', 3000),
-> (4, 'Deposit', 15000), (5, 'Withdrawal', 5000),
-> (6, 'Deposit', 4000), (7, 'Deposit', 20000),
-> (8, 'Withdrawal', 1000), (9, 'Deposit', 7000),
-> (10, 'Withdrawal', 2000), (11, 'Deposit', 12000),
-> (12, 'Deposit', 3000), (13, 'Withdrawal', 1000),
-> (14, 'Deposit', 8000), (15, 'Withdrawal', 1500),
-> (1, 'Transfer', 5000), (2, 'Transfer', 5000),
-> (3, 'Deposit', 6000), (4, 'Withdrawal', 4000);
Query OK, 20 rows affected (0.204 sec)
Records: 20 Duplicates: 0 Warnings: 0
```

```
mysql> select*from customers;
+-----+-----+-----+-----+-----+-----+
| customer_id | customer_name | email | phone | address | created_at |
+-----+-----+-----+-----+-----+-----+
| 1 | Suresh Kumar | suresh@gmail.com | 9000000001 | Bangalore | 2026-02-09 21:35:09 |
| 2 | Ananya Rao | ananya@gmail.com | 9000000002 | Mysore | 2026-02-09 21:35:09 |
| 3 | Ravi Teja | ravi@gmail.com | 9000000003 | Chennai | 2026-02-09 21:35:09 |
| 4 | Sneha Patel | sneha@gmail.com | 9000000004 | Mumbai | 2026-02-09 21:35:09 |
| 5 | Suman Das | suman@gmail.com | 9000000005 | Kolkata | 2026-02-09 21:35:09 |
| 6 | Kiran Shah | kiran@gmail.com | 9000000006 | Delhi | 2026-02-09 21:35:09 |
| 7 | Sanjay Verma | sanjay@gmail.com | 9000000007 | Pune | 2026-02-09 21:35:09 |
| 8 | Swathi N | swathi@gmail.com | 9000000008 | Bangalore | 2026-02-09 21:35:09 |
| 9 | Amit Jain | amit@gmail.com | 9000000009 | Jaipur | 2026-02-09 21:35:09 |
| 10 | Sonia Malhotra | sonia@gmail.com | 9000000010 | Delhi | 2026-02-09 21:35:09 |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.151 sec)
```

```
mysql> select*from accounts where status='active';
+-----+-----+-----+-----+-----+-----+-----+-----+
| account_id | account_number | customer_id | branch_id | account_type | balance | status | opened_date |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | ACC1001 | 1 | 1 | Savings | 25000.00 | Active | 2024-01-10 |
| 2 | ACC1002 | 1 | 2 | Current | 40000.00 | Active | 2024-02-15 |
| 3 | ACC1003 | 2 | 1 | Savings | 15000.00 | Active | 2023-12-01 |
| 4 | ACC1004 | 3 | 3 | Savings | 50000.00 | Active | 2024-03-20 |
| 5 | ACC1005 | 4 | 2 | Current | 60000.00 | Active | 2022-05-10 |
| 6 | ACC1006 | 5 | 1 | Savings | 8000.00 | Active | 2024-04-01 |
| 7 | ACC1007 | 6 | 3 | Savings | 120000.00 | Active | 2023-11-11 |
| 8 | ACC1008 | 7 | 2 | Current | 30000.00 | Active | 2024-01-25 |
| 9 | ACC1009 | 8 | 1 | Savings | 45000.00 | Active | 2024-02-05 |
| 10 | ACC1010 | 9 | 3 | Savings | 9000.00 | Active | 2023-06-15 |
| 11 | ACC1011 | 10 | 1 | Current | 70000.00 | Active | 2024-03-12 |
| 12 | ACC1012 | 2 | 2 | Savings | 20000.00 | Active | 2024-04-10 |
| 13 | ACC1013 | 3 | 1 | Current | 10000.00 | Active | 2023-09-09 |
| 14 | ACC1014 | 4 | 3 | Savings | 55000.00 | Active | 2024-01-01 |
| 15 | ACC1015 | 5 | 2 | Savings | 18000.00 | Active | 2024-02-20 |
+-----+-----+-----+-----+-----+-----+-----+-----+
15 rows in set (0.188 sec)
```

```
mysql> select*from branches where branch_id=1;
+-----+-----+-----+-----+
| branch_id | branch_name | city | IFSC_code |
+-----+-----+-----+-----+
| 1 | Main Branch | Bangalore | IFSC001 |
+-----+-----+-----+-----+
1 row in set (0.014 sec)
```

```
mysql> select customer_id,COUNT(*)
-> from accounts
-> group by customer_id
-> HAVING COUNT(*)>1;
```

customer_id	COUNT(*)
1	2
2	2
3	2
4	2
5	2

5 rows in set (0.229 sec)

```
mysql> select*from customers where
-> customer_name LIKE 'S%';
```

customer_id	customer_name	email	phone	address	created_at
1	Suresh Kumar	suresh@gmail.com	9000000001	Bangalore	2026-02-09 21:35:09
4	Sneha Patel	sneha@gmail.com	9000000004	Mumbai	2026-02-09 21:35:09
5	Suman Das	suman@gmail.com	9000000005	Kolkata	2026-02-09 21:35:09
7	Sanjay Verma	sanjay@gmail.com	9000000007	Pune	2026-02-09 21:35:09
8	Swathi N	swathi@gmail.com	9000000008	Bangalore	2026-02-09 21:35:09
10	Sonia Malhotra	sonia@gmail.com	9000000010	Delhi	2026-02-09 21:35:09

6 rows in set (0.017 sec)

```
mysql> select*from accounts where
-> balance between 10000 and 50000;
```

account_id	account_number	customer_id	branch_id	account_type	balance	status	opened_date
1	ACC1001	1	1	Savings	25000.00	Active	2024-01-10
2	ACC1002	1	2	Current	40000.00	Active	2024-02-15
3	ACC1003	2	1	Savings	15000.00	Active	2023-12-01
4	ACC1004	3	3	Savings	50000.00	Active	2024-03-20
8	ACC1008	7	2	Current	30000.00	Active	2024-01-25
9	ACC1009	8	1	Savings	45000.00	Active	2024-02-05
12	ACC1012	2	2	Savings	20000.00	Active	2024-04-10
13	ACC1013	3	1	Current	10000.00	Active	2023-09-09
15	ACC1015	5	2	Savings	18000.00	Active	2024-02-20

9 rows in set (0.026 sec)

```
mysql> select opened_date,COUNT(*)
-> from accounts
-> group by opened_date
-> HAVING COUNT(*)>2024;
Empty set (0.197 sec)
```

```
mysql> select*from accounts
-> where status ='closed';
Empty set (0.013 sec)
```

```
mysql> select branch_id ,sum(balance) from accounts group by branch_id;
```

branch_id	sum(balance)
1	173000.00
2	168000.00
3	234000.00

```
3 rows in set (0.014 sec)
```



```
mysql> select avg(balance) from accounts;
```

avg(balance)
38333.333333

```
1 row in set (0.152 sec)
```

```
mysql> select customer_id,balance
-> from accounts
-> order by balance DESC;
```

customer_id	balance
6	120000.00
10	70000.00
4	60000.00
4	55000.00
3	50000.00
8	45000.00
1	40000.00
7	30000.00
1	25000.00
2	20000.00
5	18000.00
2	15000.00
3	10000.00
9	9000.00
5	8000.00

```
15 rows in set (0.017 sec)
```



```
mysql> select customer_id,balance  
-> from accounts  
-> order by balance DESC;
```

customer_id	balance
6	120000.00
10	70000.00
4	60000.00
4	55000.00
3	50000.00
8	45000.00
1	40000.00
7	30000.00
1	25000.00
2	20000.00
5	18000.00
2	15000.00
3	10000.00
9	9000.00
5	8000.00

```
15 rows in set (0.016 sec)
```

```
mysql> select customer_id,balance
-> from accounts
-> order by balance DESC;
```

customer_id	balance
6	120000.00
10	70000.00
4	60000.00
4	55000.00
3	50000.00
8	45000.00
1	40000.00
7	30000.00
1	25000.00
2	20000.00
5	18000.00
2	15000.00
3	10000.00
9	9000.00
5	8000.00

15 rows in set (0.016 sec)

```
mysql> select customer_id,balance
-> from accounts
-> order by balance DESC
-> LIMIT 1;
```

customer_id	balance
6	120000.00

1 row in set (0.013 sec)

```
mysql> select branch_id ,COUNT(*)  
-> from accounts  
-> group by branch_id;
```

branch_id	COUNT(*)
1	6
2	5
3	4

3 rows in set (0.015 sec)